



Meridian Junior College
JC2 Preliminary
Examinations

Name:

Class:

Index No:

BIOLOGY

9747/02

Paper 2 Core Paper

08 Sep 2008

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and register number on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination,

1. Write your name, index number and civics group on all sheets of writing papers.
2. Fasten all your writing papers together.
3. Hand in Section A and Section B separately.
4. Write down the question number for the question answered for Section B.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
Total	

Section A

Answer **all** questions in this section.

1. Fig 1.1 is an electron micrograph of a plasma cell from bone marrow.

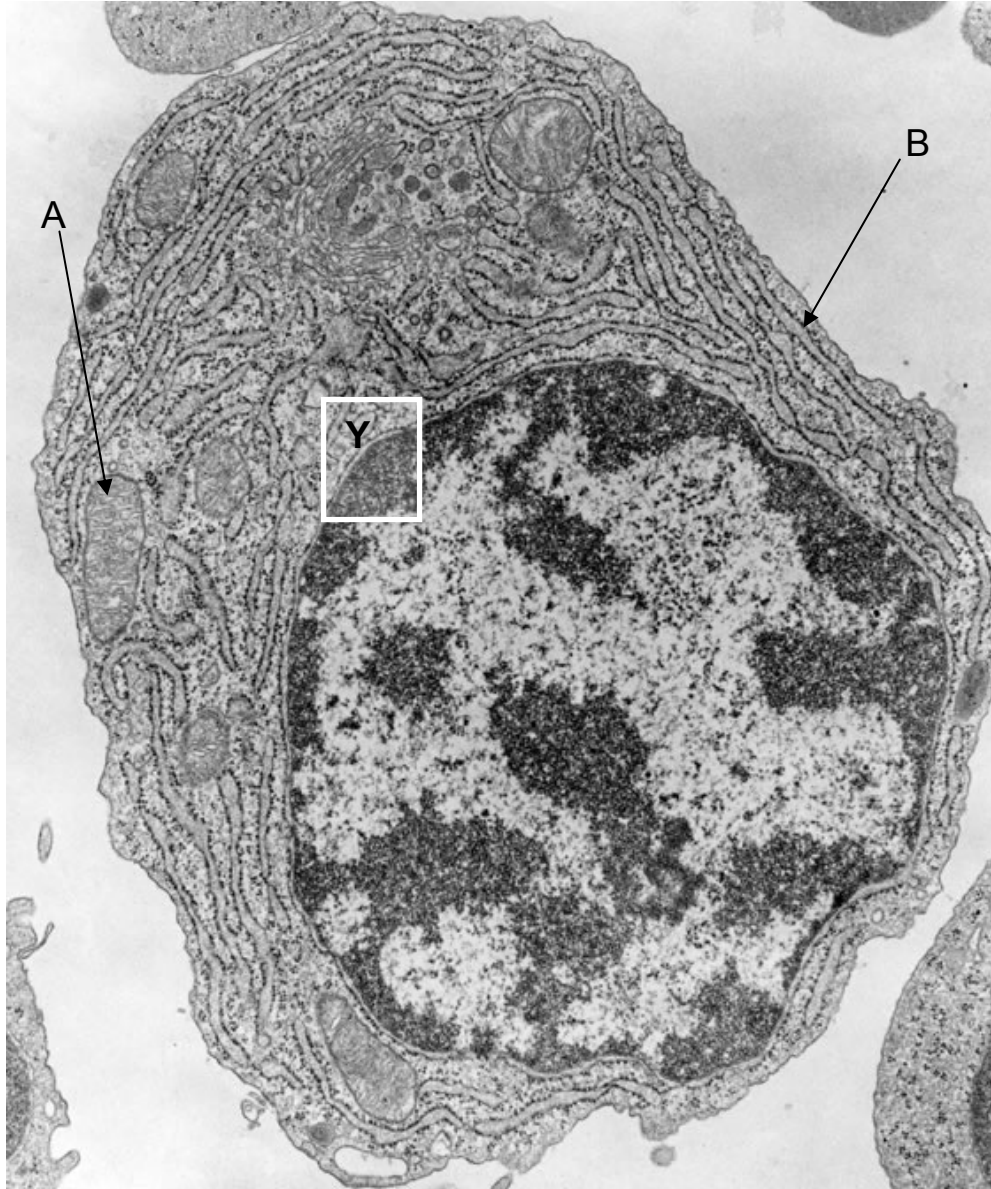


Fig. 1.1

(a) Identify organelles A and B.

[2]

- **A is mitochondrion**
- **B is rough endoplasmic reticulum.**

(b) With reference to Fig. 1.1, briefly describe the structural features of organelle B.

[2]

- It consists of **cisternae**, which are flattened and interconnected with each other and are **continuous with the outer membrane of the nuclear envelope**.
- It bears **ribosomes** on the **surface** for protein synthesis.

Fig. 1.2 shows a diagrammatic representation of Region Y in Fig. 1.1.

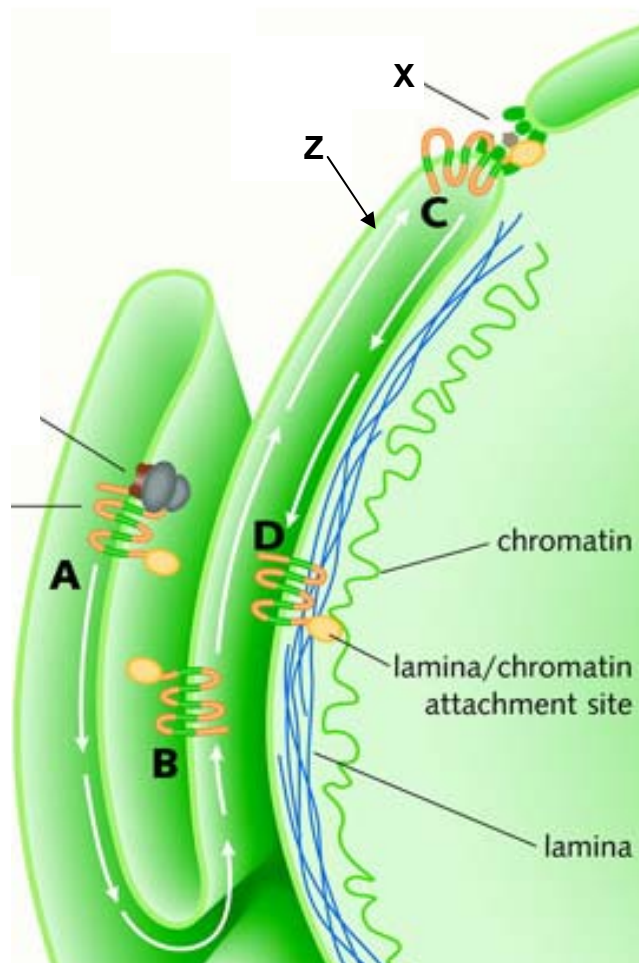


Fig. 1.2

(c) Briefly describe the role of X.

[1]

- It allows processed **mRNA to exit** the **nucleus** / allows **nucleotides to enter** the **nucleus**.

(d) Describe the structure in Region Z.

[2]

For
Examiner's
Use only

- It is made up of a **bilayer of phospholipids**,
- where they are arranged such that the **hydrophilic phosphate heads** are facing external and internal **aqueous environment** while **hydrophobic hydrocarbon tails** are **kept within bilayer**.

Fig. 1.3 shows a section of the cell membrane.

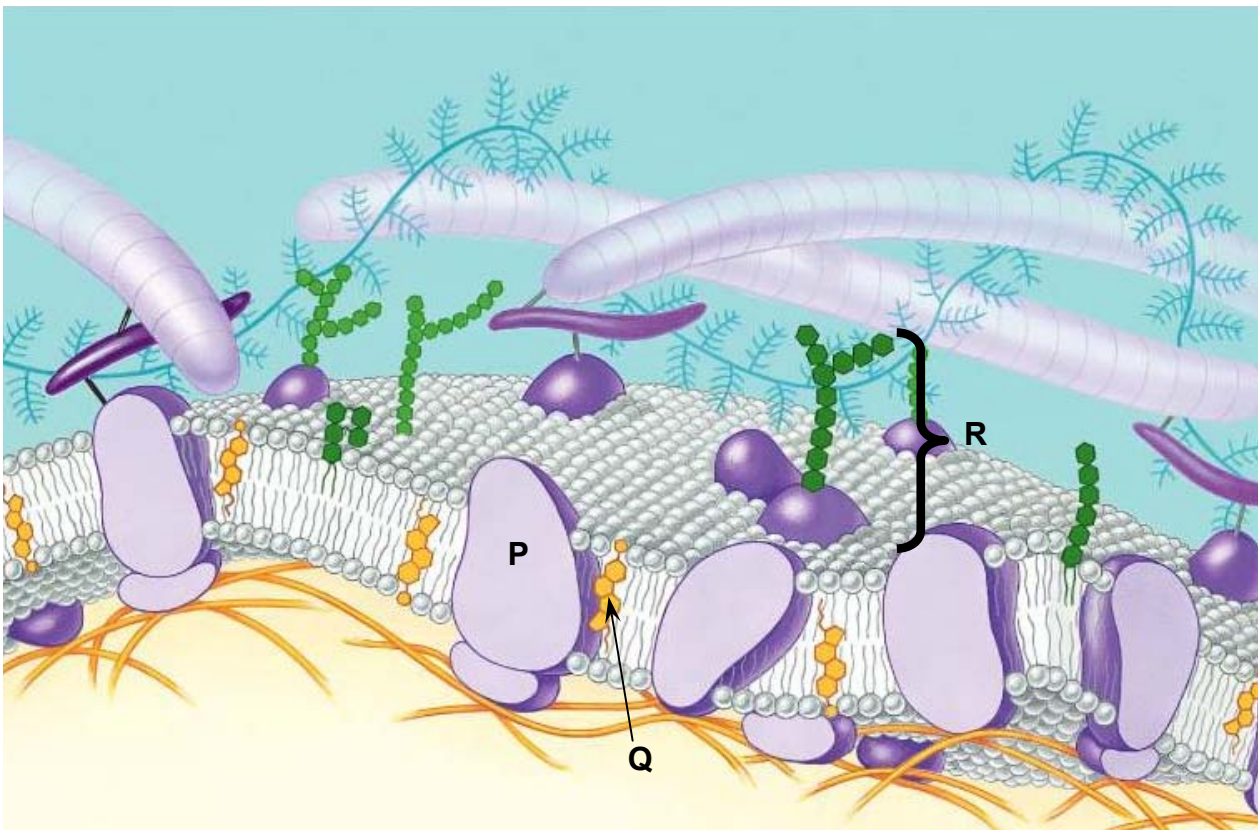


Fig. 1.3

(e) With reference to Fig. 1.3, identify the labelled structures and state their roles.

[3]

P: intrinsic / integral / transmembrane protein – involved in **transport** of molecules across the membrane/ **carrier proteins/ channels/ pumps/** serves as **enzyme** for biochemical or metabolic **reactions/ receptor for stimuli** leading to signal transduction/ for **intracellular joining/ attachment to cytoskeleton**.

Q: cholesterol – **regulating fluidity** of the **cell membrane**.

R: glycoprotein – involved in **cell to cell recognition / cell adhesion**.

(f) State two possible functions of cell membrane.

[2]

- It enables **compartmentalization of the cell**/ acts as a barrier to **separate the contents of the cells from the external environment**.
- It **controls** the **exchange of substances** between **cell and external environment**.
- It **consists of receptor sites** for recognizing external stimuli such as hormones.

OR

It enables **cell-cell recognition**.

OR

It enables **enzyme** activity for biochemical or metabolic **reactions**.

OR

It enables **intercellular joining**.

OR

It enables **attachment to cytoskeleton and the extracellular matrix**.

[Any 2]

[12 Marks]

2. Fig. 2.1 is a diagram showing the changes in membrane potential of a nerve cell.

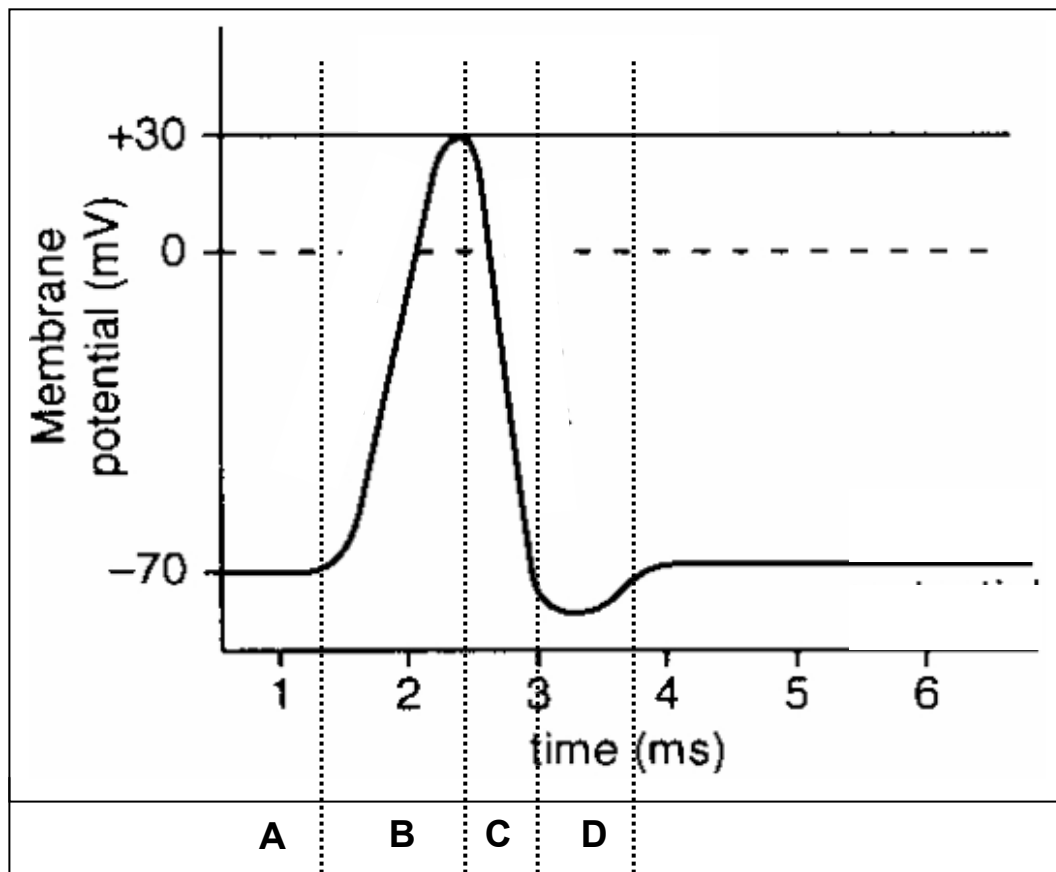


Fig. 2.1

(a) Identify the stages **A** to **D** of an action potential.

[2]

- A:** Resting state
- B:** Depolarisation
- C:** Repolarisation
- D:** Hyperpolarisation

0-1 correct: 0 mark

2-3 correct: 1 mark

4 correct: 2 marks

(b) With reference to Fig. 2.1, explain Stages B and C.

[4]

- In **Stage B**, there is an **increase in membrane potential from -70mV to +30mV**.
- This is due to the **opening of the gated sodium channels**, which causes an **influx of sodium ions**, thus increasing the membrane potential.
- In **Stage C**, there is a **decrease in membrane potential from +30mV to -70mV**.
- This is due to the **closing of the sodium gated channels** as well as the **opening of the gated potassium channels**, which lead to an **efflux of potassium ions**.

Fig. 2.2 shows a molecular structure of tetrodotoxin, a potent neurotoxin with no known antidote. The binding site of this toxin is located at the pore opening of the voltage-gated Na^+ channel.

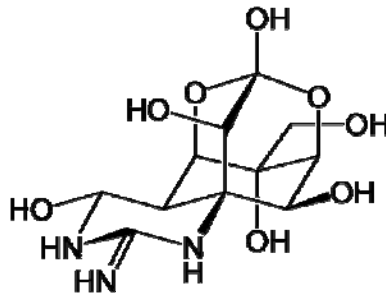


Fig. 2.2

(c) Suggest why tetrodotoxin is considered a lethal neurotoxin.

[3]

- **Binding of toxin will block the opening of the sodium-gated channels.**
- This will **prevents sodium influx** and **no action potential / depolarisation can occur**.
- Therefore, **neurotransmission stops**.
- This can lead to **muscular paralysis / respiratory depression / circulatory failure**.

[Any 3]

Some bacteria produce tetrodotoxin, which can be found in the internal organs of the pufferfish, as they share a symbiotic relationship. This makes them a possibly lethal meal for any predatory fish. Tetrodotoxin is found primarily in the ovaries and liver, although smaller amounts exist in the intestines and skin, as well as trace amounts in muscle tissue and blood.

The pufferfish has immunity to the poison due to a mutation in the protein sequence of the sodium-gated channel on the cell membranes.

(d) Suggest how this mutation confers immunity to tetrodotoxin in pufferfish. [1]

- Mutation leads to **conformation change in the sodium-gated channel** and **prevents the binding of tetrodotoxin / binding site is relocated** away from pore opening.

OR

- **No blockage** by tetrodotoxin occurs and **action potential / depolarisation can be generated** for transmission of impulses.

[10 Marks]

3.

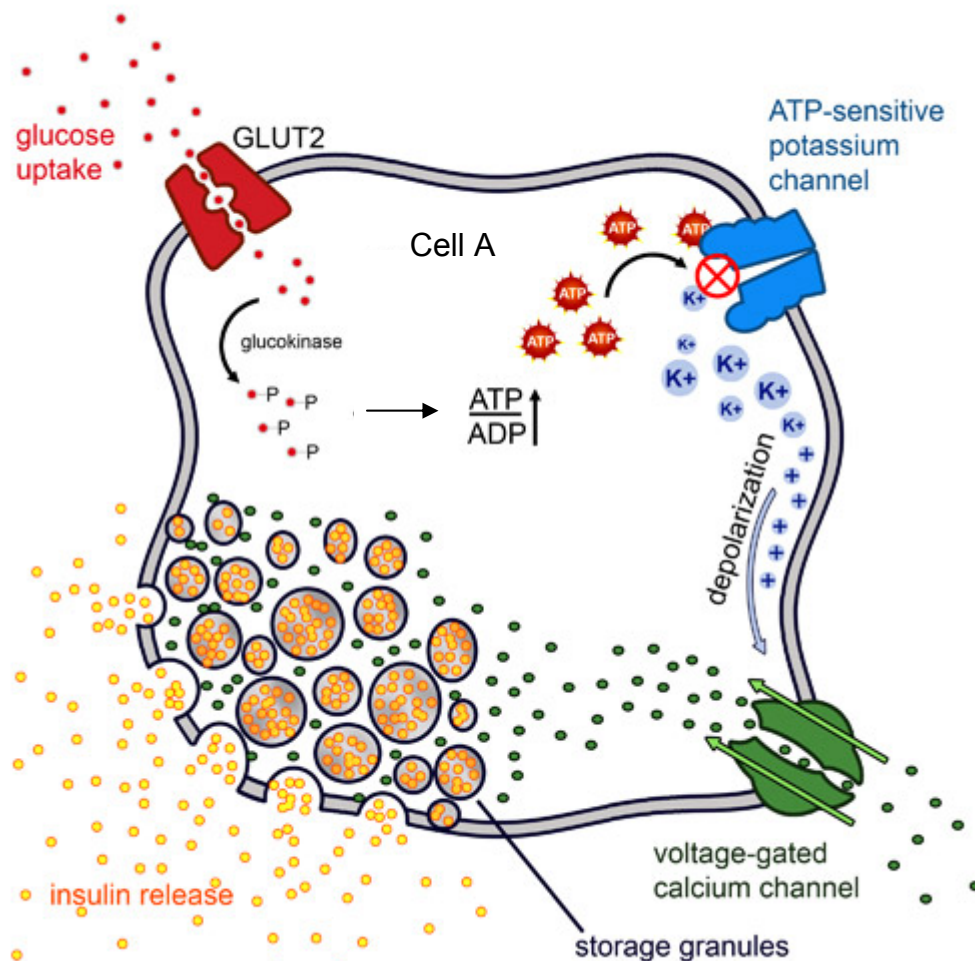


Fig. 3.1

Fig. 3.1 shows that insulin secretion in Cell A is triggered by rising blood glucose levels. The uptake of glucose causes a rise in the ATP:ADP ratio in the cell. This in turn closes the potassium channel and leads to depolarization of the membrane, causing the voltage-gated calcium channel to open. The influx of calcium ions leads to the exocytotic release of insulin from their storage granules.

(a) Identify Cell A [1]

- **β cell of islets of Langerhans** in pancreas.

(b) Explain how GLUT2 allows the passage of glucose. [1]

- GLUT2 is a protein spanning the membrane and provides a hydrophilic **channel**/act as **carrier protein** across the membrane that is selective for glucose.

- (c) Explain how dysfunctional GLUT2 in diabetic patients leads to their required regular injections of insulin. [3]
- Cell A of patients with dysfunctional **GLUT2 does not allow the diffusion/influx/uptake of glucose** through the cell membrane of β cells of islets of Langerhans.
 - With the glucose molecules failing to diffuse through the cell membrane, **insulin is not secreted** by the β cells of islets of Langerhans.
 - This makes the β cells of islets of Langerhans of patients unresponsive to high blood glucose levels, thereby **requiring injection of insulin to lower blood glucose level to the norm**.
- (d) Suggest the reason for the increase of ATP:ADP ratio following the uptake of glucose through GLUT2. [2]
- With the increased uptake of glucose, the **rate of aerobic respiration/glycolysis increases** that leads to the **increased production of ATP**.
 - The **decrease in ADP concentration** due to the **phosphorylation of ADP into ATP/(through) substrate level phosphorylation/oxidative phosphorylation**.
- (e) Explain why insulin is defined as an endocrine hormone. [1]
- Insulin secreted by the β cells of the Islets of Langerhans of the pancreas is **secreted into the blood stream** to distal target/liver/muscle cells.
- (f) Outline the action of insulin in the signaling pathway upon reaching the liver and muscle cells. [4]
- **Insulin binds** to the insulin receptor which is a class of **tyrosine-kinase receptors**.
 - This leads to the **dimerisation of the tyrosine kinase receptors**.
 - This **activates** the **tyrosine-kinase** sections of both which **add phosphates** to the **tyrosine tails** of the other polypeptide.
 - The phosphorylated tyrosine kinase tails **activate** a variety of specific **relay proteins** by binding.
 - This leads to **glucose transporter (Glut4) transported from vesicles to cell surface membrane** and **mediate the transport of glucose into the liver/muscle cells**.

[Max 4]

[12 Marks]

4. Fig. 4.1 shows a cell undergoing two different stages of meiosis (to the same scale).

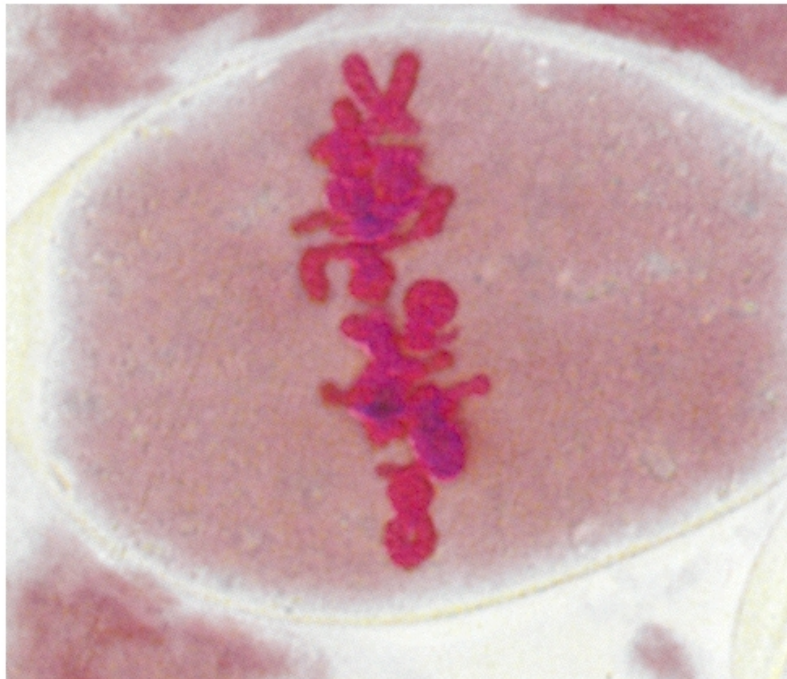
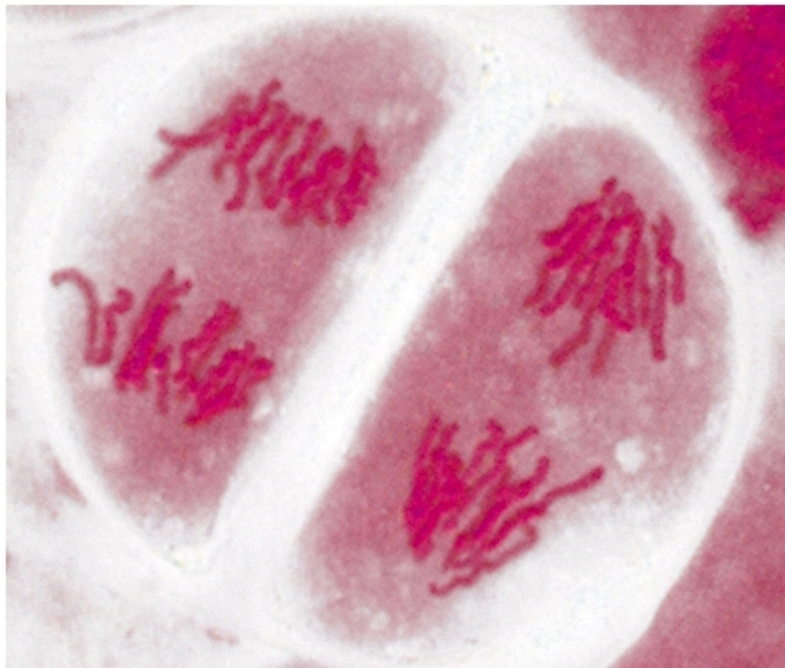
**A****B**

Fig. 4.1

(a) Identify stages **A** and **B**.

[2]

- A is **metaphase I**
- B is **anaphase II**.

(b) Explain the phenomenon observed in **B**. [2]

- **Centromeres divide**
- and the **chromatids separate** and are pulled, centromeres first, by the spindle fibres **to opposite poles**.
- This is due to the **shortening of** the centromere-to-pole **spindle fibres**.

[Any 2]

(c) Describe how meiosis can lead to genetic variation. [2]

- In **Prophase I**, non-sister chromatids of **homologous chromosomes** align themselves alongside one another, and **form chiasma**. **Crossing over** occurs and new combinations of alleles on the chromosomes of the gametes are formed.
- During **Metaphase I** and **Anaphase I** respectively, the **arrangement and subsequent separation of the chromosomes** of each bivalent is completely **independent of** the orientation of the **other bivalents**. This gives rise to different combinations of chromosomes in a gamete.
- During **Metaphase II** and **Anaphase II** respectively, the **arrangement and subsequent separation of the chromatids** of each chromosome is **completely independent of** the orientation of the **other chromosomes**. This gives rise to different combinations of chromosomes in a gamete.

[Any 2]

Okadaic acid is a toxin that causes spindle fibres to significantly increase in length and disrupts the interactions between microtubules and kinetochores, resulting in a disorganized metaphase plate.

(d) Suggest a possible outcome of a disorganised metaphase plate. [2]

- The **chromosomes and chromatids may not separate equally/properly** during **Anaphase I or II** respectively.
- **Meiosis cannot continue** and **no germ cells** are formed / This lead to **non-disjunction**.

OR

- The **sister chromatids** may not separate equally/properly during **Anaphase**.
- **Mitosis / cell division cannot continue** and **no daughter cells** are form / This leads to **non-disjunction / Aneuploidy / Polyploidy**.

[8 Marks]

5. Hunter's syndrome is a serious genetic disorder and it usually affects males. It interferes with the body's ability to break down and recycle specific mucopolysaccharides, also known as glycosaminoglycans or GAG. The visible signs and symptoms of Hunter syndrome in younger people are usually the first clues leading to a diagnosis. In general, the time of diagnosis usually occurs from about 2 to 4 years of age.

Fig. 5.1 shows the pedigree of Mariah's family. Mariah (M) married Nick (N) and had three children. One of the children had Hunter's syndrome.

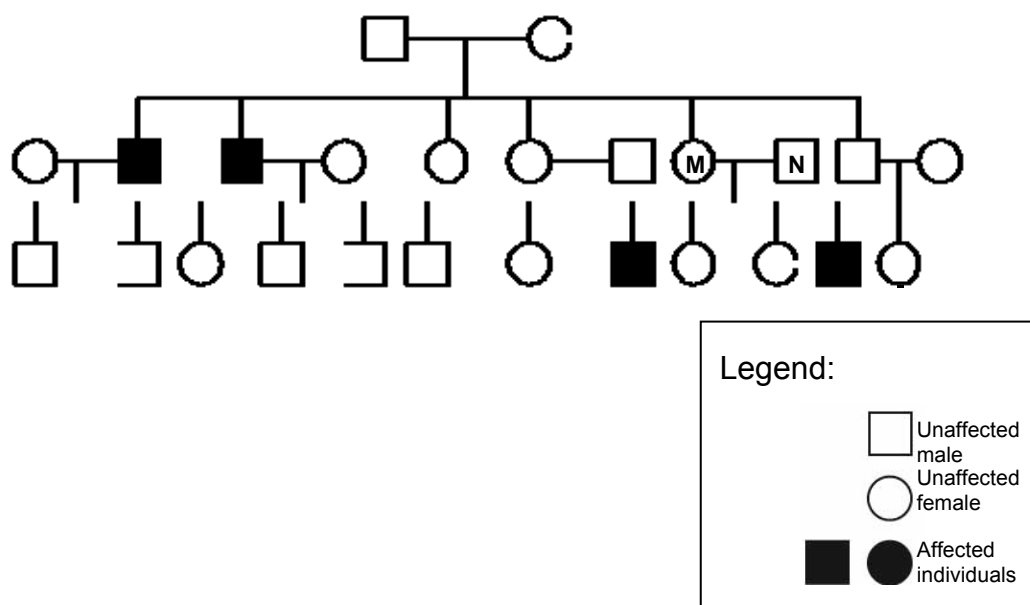


Fig. 5.1

- (a) With reference to Fig. 5.1, determine the type of inheritance for Hunter's syndrome. [1]

- **Sex-linked** inheritance.

(b) Using a genetic diagram, explain the type of inheritance pattern seen in the offspring of Mariah and Nick. [6]

Let H be the allele for normal and h be the allele for Hunter's syndrome, where H is dominant over h .

Parental phenotypes	Normal Female	X	Normal Male	
Parental genotypes (2n)	$X^H X^h$	X	$X^H Y$	
Gametes (n)	X^H X^h	X	X^H Y	
F ₁ genotypes (2n)	$X^H X^h$ $X^H Y$		$X^H X^H$ $X^h Y$	
F ₁ phenotypes	Normal Female	Normal Male	Normal Female	Hunter's Syndrome Male
F ₁ phenotypic ratio	2 Normal Female: 1 Normal Male: 1 Hunter's Syndrome Male			

A survey was conducted on the occurrence of Hunter's syndrome in Singapore. Information was gathered from 100 couples whereby the female is a carrier for Hunter's syndrome, while the male is normal. Only the first child of each couple was tested for Hunter's syndrome. In total, it was found that 6 children had the disease.

(c) Use a suitable test with a 5% level of significance to determine whether the inheritance of the disease in the survey follow Mendelian's pattern of inheritance. [6]

- χ^2 test is used.
- **Null Hypothesis:** There is **no significant difference** between the **observed occurrence** of children with **Hunter's syndrome** and the **expected occurrence ratio of 1:3**.

Alternative Hypothesis: There is a **significant difference** between the **observed occurrence** of children with **Hunter's syndrome** and the **expected occurrence ratio of 1:3**.

Category	Observed Frequency (O)	Expected Frequency (E)	(O-E)	$\frac{(O-E)^2}{E}$
Normal	94	$3/4 \times 100 = 75$	19	4.813
Hunter's Syndrome	6	$1/4 \times 100 = 25$	-19	14.440

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

$$= 4.813 + 14.440$$

$$= \mathbf{19.25 \text{ (2 d.p)}}$$

- **Degree of Freedom, $\nu = 2-1 = 1$**
- From the χ^2 table,
At 5 % level of significance and $\nu = 1$,
 $\chi^2_{\text{critical}} = \mathbf{3.84}$
- As $\chi^2_{\text{calculated}} > \chi^2_{\text{critical}}$, the **Null hypothesis is rejected** and the **Alternative hypothesis is accepted**. There is a significant difference between observed occurrence of children with Hunter's syndrome and the expected occurrence ratio of 1:3.
- **The pattern of inheritance in survey do not follow Mendelian's pattern of inheritance.**

[Any 6]

(d) Suggest one possible reason why the survey conducted may not be a fair one.

[1]

- The **sample size is too small** / only the **1st child** of the couple **was accounted** in the survey / **gender distribution may not be equal**.

[14 Marks]

6. In 1975 a team of Japanese scientists discovered a strain of *Flavobacterium* living in ponds containing waste water from a factory producing nylon. *Flavobacterium* are capable of digesting certain by-products of nylon, even though those substances are not known to have existed before the invention of nylon in 1935. This strain of *Flavobacterium* sp. K172, became popularly known as *nylon-eating bacteria*, and the enzymes are collectively known as *nylonase*. *Nylonase* were found to have no catalytic activity towards naturally occurring amide compounds, suggesting that the enzymes are completely new, as there is no homology in the amino acid sequence found between *nylonase* and other known enzymes.

Other species of *Flavobacterium* are known to commonly cause diseases in freshwater fishes as they derive their carbon and nitrogen sources from them. Due to the poisoning by factory wastes, the population of fish in the pond has declined.

- (a) It was suggested that the gene for coding *nylonase* may have arisen as a result of mutation. Suggest and explain the type of gene mutation that could have taken place. [2]
- **Frameshift mutation/Insertion/Deletion** of nitrogenous bases.
 - There are **no homologies** found with known **enzymes**, suggesting frameshift mutation/insertion/deletion/**new protein** is formed. OR
 - **Substitution is improbable** as **most of the amino acid sequence remains intact**.
- (b) Account for the predominance of *nylon-eating bacteria* in the ponds through natural selection. [3]
- The **selection pressure** in the ponds is the **lack of carbon and nitrogen sources/lack of food** for the *Flavobacterium* as the population of fishes declined in numbers.
 - There is **variation** in strains of *Flavobacterium* with K172/some existing *Flavobacterium* being **able to break down nylon/contains the gene coding for nylonase**.
 - The ***nylon-eating bacteria* are at selective advantage** as they **survive to reproduce, passing down** the advantageous **traits/genes/alleles** down to the subsequent generations.

(c) It is unlikely for *nylon-eating bacteria* to be commonly found in freshwaters that are unpolluted. Explain why. [2]

- *Nylon-eating bacteria* are **not at selective advantage compared to other *Flavobacterium***.
- This is because the **selective pressure is absent/different**.

(d) Suggest how one can estimate quantitatively the phylogenetic distance between two evolutionarily related bacteria. [2]

- The **DNA sequence/amino acid sequence** of the bacteria are **compared**.
- The **more regions of homology** they have, the **more related they are**./The **less regions of homology** they have, the **less related they are**.

OR

- The **DNA** of both bacteria are isolated and **denatured to give single DNA strands** and mixed and analysed.
- The **more complementary base pairs** they have, the **more related with are**./ The **less complementary base pairs** they have, the **less related with are**.

(e) Ecological speciation concept is generally preferred over biological speciation concept in classifying bacteria such as *Flavobacterium*. Explain why. [1]

- **Ecological speciation concept accommodates asexual species** whereas the **biological species requires organisms to be reproducing through sexual means** which does not accommodate *Flavobacterium*.

(f) The classification of *Flavobacterium gondwanense* is listed as shown below in decreasing levels of taxa. Fill in the blanks. [2]

Levels of taxa	Name of Taxa
Kingdom	Bacteria
▪ Phylum	Bacterioidetes
Class	Flavobacteria
▪ Order	Flavobacteriaceae
Family	Flavobacteriaceae
Genus	▪ <u>Flavobacterium</u>
Species	▪ <u>Gondwanese</u>

(0-1 correct = 0 mark)

(2-3 correct = 1 mark)

(4 correct = 2 marks)

[12 Marks]

7. The *lac* operon consists of three structural genes, a promoter, a terminator, a regulator, and an operator as shown in Fig. 7.1. The three structural genes are: *lacZ*, *lacY*, and *lacA*. *LacZ* encodes β -galactosidase, an intracellular enzyme that cleaves the disaccharide lactose into glucose and galactose. *LacY* encodes β -galactoside permease, a membrane-bound transport protein that pumps lactose into the cell. *LacA* encodes β -galactoside transacetylase, an enzyme that transfers an acetyl group from acetyl-CoA to β -galactosides.

Only *lacZ* and *lacY* appear to be necessary for lactose catabolism. The *lacI* shown in Fig. 7.1 is sometimes also considered as part of the *lac* operon.

Bacteria usually prefer to use glucose over lactose as source of energy.

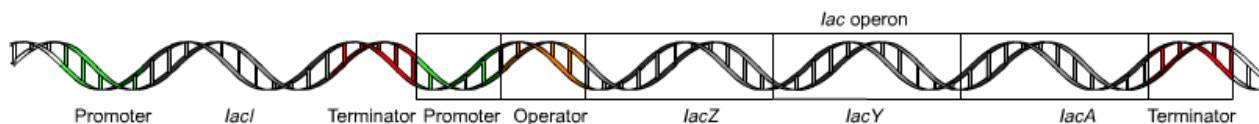


Fig. 7.1

- (a) Explain why bacteria prefer glucose over lactose as energy source. [2]
- **Glucose is monosaccharide** while **lactose is disaccharide**.
 - It takes more energy for bacteria to **hydrolyse lactose into monosaccharide** before it can use **as an energy source/respiration**. OR
 - Bacteria can **use glucose directly for respiration/energy** source. OR
 - **Easier to metabolize / oxidize / breakdown glucose** to produce energy.
- (b) Describe how the presence of allolactose turns the expression of the *lac* operon on. [2]
- Allolactose **binds to the repressor** molecule and **causes conformational change** to/**inactivates** the repressor molecule.
 - **Repressor dislodges from/cannot bind** to the **operator** region of the operon and **allows RNA polymerase to transcribe** the structural **genes**.
- (c) State the function of the following in the regulation of *lac* operon expression. [1]
- (i) Promoter
- This is the **binding site** for the **RNA polymerase**.

(ii) *LacI*

[1]

- This is the gene **coding** for the **repressor molecule**.

Fig. 7.2 shows the polycistronic nature of lac operon.

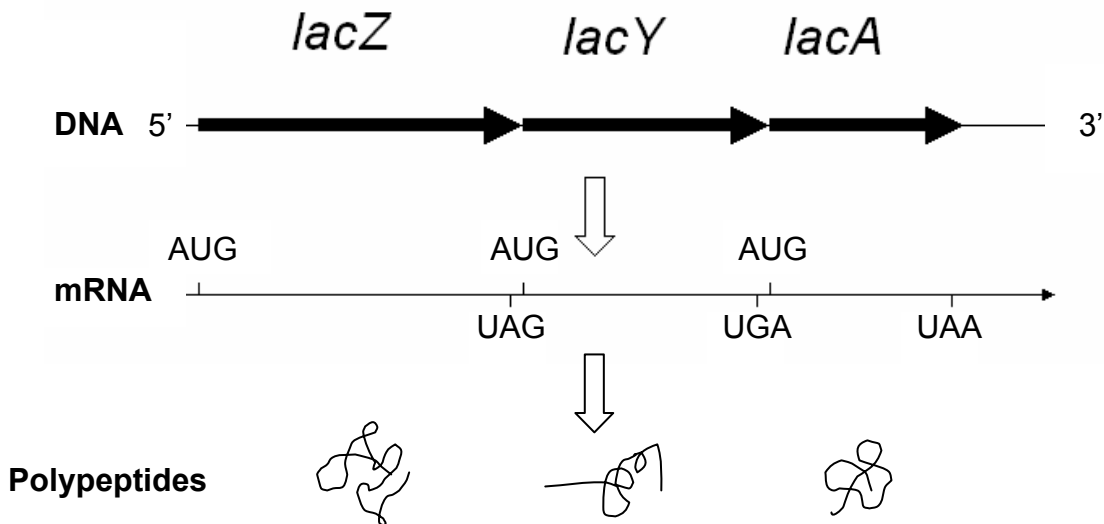


Fig. 7.2

(d) Explain how Fig. 7.2 illustrates the polycistronic nature of the lac operon. [2]

- Upon transcription, the mRNA for the *lac* operon contains **three start codons** and **three stop codons**.
- This gives rise to a total of **three different polypeptides** can be **translated from / encoded by the mRNA**, illustrating the polycistronic nature of *lac* operon.

A certain bacteriophage is known to insert its own genome between the *lacZ* and *lacY* structural genes to give prophages.

(e) Explain the change in number of bacteriophage (if any) in conditions when host bacteria are grown in environment with

(i) abundant supply of glucose and no lactose supply. [2]

- The bacteriophage remains in the **lysogenic cycle**.
- This is because the **operon is not turned on** due to the absence of lactose / **no transcription of genes**.
- Therefore, the **viral proteins are not produced** to form viruses/**no change in the number of bacteriophage**.

[Max 2]

(ii) no supply of glucose and abundant supply of lactose.

[2]

- The bacteriophage **switch to lytic cycle**.
- This is because the **operon is turned on** due to the presence of lactose and absence of glucose / **transcription of genes**.
- Therefore, the **viral proteins are produced** to form viruses/**increase in the number of bacteriophage**.

[Max 2]

[12 Marks]

Section B

Answer **one** question.

Write your answers on a separate answer paper provided.

Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answers should be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated by the question.

8. (a) Outline the main stages of Calvin Cycle. [6]

1. The first stage of Calvin Cycle is carbon fixation, which involves the incorporation one molecule of **carbon dioxide** to a 5C **ribulose biphosphate** (RuBP) to give a **6C unstable intermediate**.
2. This is **catalysed** by the enzyme **ribulose biphosphate carboxylase oxygenase / (RUBISCO)**.
3. This **6C intermediate** is **unstable**; hence it is **broken down** to give two molecules of 3C **glycerate-3-phosphate** (GP), the first product.
4. The second stage is carbon reduction, where **glycerate-3-phosphate** is **reduced** to a 3C **triose phosphate** (TP) / **Glyceraldehyde-3-phosphate** (GALP),
5. in the presence of **ATP** and **reduced NADP / NADPH**.
6. The last stage is ribulose biphosphate regeneration. Some **triose phosphates** are used to **regenerate RuBP**,
7. using **ATP** from the light dependent stage to **rearrange the carbon atoms** between sugar phosphates to generate a 5C ribulose biphosphate from 3C sugar.
8. For **every 6 molecules of triose phosphate** formed, **1 molecule of triose phosphate exits** to be converted to glucose, amino acids and lipids.

[Any 6]

(b) Explain the importance of oxidative phosphorylation. [8]

1. It occurs during **aerobic respiration**.
2. Oxidative phosphorylation **produces 90% of the ATP** generated by respiration / **34 out of 38 ATPs**.

3. As **electrons are passed down the Electron Transport Chain**, **energy released** is used to **create a proton concentration gradient**, across the inner membrane of mitochondria, needed for ATP synthesis.
4. This **creates the proton motive force**.
5. Hydrogen ions are passed back into the mitochondrial matrix, down a concentration gradient, through **ATP synthase** to **synthesize ATP**.
6. **Each reduced NAD** molecule potentially produced **three molecules of ATP**.
7. **Each reduced FAD** molecule potentially produced **two molecules of ATP**.
8. Oxidative phosphorylation also allows the **oxidation of NADH and FADH₂** / to **regenerate NAD⁺ and FAD**.
9. **Oxygen** act as a **final electron acceptor** in the Electron Transport Chain and **maintain the flow of electrons along the Electron Transport Chain**.
10. **Metabolic water** is also produced through **reduction of O₂** / when the **electron** is finally **transferred to oxygen**.

[Any 8]

- (c) Explain what happens in an animal cell when there is insufficient oxygen for aerobic respiration. [6]

1. The animal cell will undergo **anaerobic respiration / lactate fermentation**.
2. **NADH** from glycolysis **reduces pyruvate to lactate / lactic acid**, forming NAD⁺.
3. **NAD⁺ is regenerated for glycolysis**.
4. Glycolysis **produces 2 net ATP** from every molecule of Glucose.
5. **Lactic acid** in **blood** and **muscle** build up and has to be removed because it is **toxic** to cells at **high concentration**.
6. When **oxygen is available again**, the liver **oxidizes lactate** to **carbon dioxide** and **water** via **aerobic respiration / pyruvate respire to give CO₂ + H₂O**
7. **lactate** can also be **converted to pyruvate / glycogen** by the liver.

[Any 6]

- 9 (a) Describe the process of transcription in prokaryotes. [6]
1. **RNA polymerase binds to the promoter region/TATA** of the gene.
 2. This **separates the double-helix of DNA strands** at the appropriate point unwinding 20 bases at a time.
 3. **RNA polymerase initiate** transcription by **base pairing ribonucleotides** with its **complementary DNA bases** on the sense strand through hydrogen bonds.
 4. **RNA polymerase** will also **catalyse the formation** of the **phosphodiester bonds** between adjacent ribonucleotides.
 5. RNA polymerase adds **ribonucleotides** to the **3' end** of the growing polynucleotide chain/**genes on the DNA are read in the 3' to 5' direction/** creating a **5' to 3' mRNA strand**.
 6. 5' to 3' elongation of the mRNA strand occurs till the **terminator** (AAUAAA) is reached.
 7. **RNA polymerase continues a bit more beyond** the terminator sequence (AAUAAA) before it dissociates from mRNA-DNA.
 8. **mRNA is released into the cytoplasm.**
 9. **DNA re-coils to its double-helix** form.

[Max 6]

- 9 (b) Explain how the amount of proteins in a eukaryotic cell can be controlled. [8]

DNA Packing

1. Genes can be packed in the **euchromatin for the genes to be transcribed/turned on/switched on** or the **heterochromatin for the genes not to be transcribed/to be turned off/to be switched off**.

Transcriptional Stage

2. To turn up the levels of transcription, there can also be **duplication of genes in the DNA**.
3. The expression of genes can be controlled by proximal control elements such as the **promoter** that **enables transcription**.
4. **Transcription factors** bind to promoter/TATA to **enhance the binding of RNA polymerase** which **increase gene expression**.
5. The expression of genes can also be controlled by distal control elements such as the **enhancers that increase the gene expression when bound to activators**.

6. The expression of genes can also be controlled by distal control elements such as the **silencers that decrease/turn off the gene expression when bound to repressors**.

Post Transcriptional stage

7. Addition of **poly A tail** at the **3' end** of the mRNA and a **5' modified guanosine cap** **stabilises the mRNA/facilitates in binding to a ribosome** (only for 5')/**allows exit of mRNA through nuclear pore** (only 3' tail).

Translational stage

8. Formation of **polyribosomes** in translation that **increases the rate of translation**.
9. Translation of specific mRNAs can be blocked by **regulatory proteins that bind to specific sequences or structures within the 5' leader region of mRNA that prevents attachment to ribosomes**.
10. **Protein factors** required to **initiate translation** in eukaryotes offer targets for simultaneously controlling translation of all the mRNA in a cell.

Post Translational stage

11. Polypeptide can also be **ubiquitinated** to label the **polypeptide for degradation** by proteosomes.

[Max 8]

- 9 (c) Describe and explain the features essential in maintaining the structure of DNA double helix. [6]

1. The **3'OH** and the **5' phosphate** in deoxyribonucleotides are important for the **formation of phosphodiester bonds** between adjacent deoxyribonucleotides.
2. The **phosphodiester bonds/sugar phosphate backbones** are important **keeping the sequence of deoxyribonucleotides in place**.
3. The **antiparallel** nature of DNA double helix aligns the DNA strands so that **hydrogen bonds can form properly** between the nitrogenous bases from opposite strands.
4. **3 H bonds between G and C and 2 H bonds between A and T** are to **provide structural stability to the DNA double helix**.
5. The **nitrogenous bases** are important for the formation of hydrogen bonds between opposite deoxyribonucleotides essential for **complementary base pairing** so that DNA can be replicated with little errors.

6. The **base pairing** is such that it takes place **between a purine and a pyrimidine** so as to **maintain the diameter of DNA at 20 Å**.
7. The DNA double helix is wound in such that it consists of **major grooves and minor grooves** alternating along the axis of DNA that is essential for the **binding with proteins/DNA polymerases/RNA polymerases**.
8. Distance between adjacent deoxyribonucleotides is 3.4 Å and with **10 deoxyribonucleotides per turn** so that the **distance between each successive turn is maintained at 34 Å**.

[Max 6]

For point 3

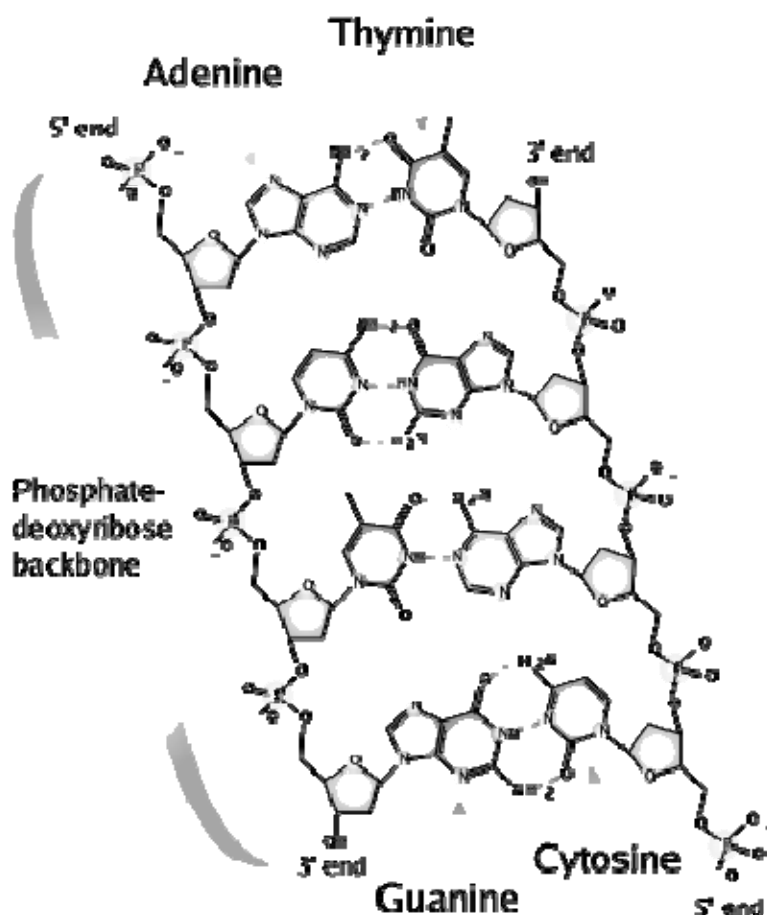


Table 5: Critical Values for the Chi-Square (χ^2) Test

		Level of significance (P)					
		0.05	0.025	0.01	0.005	0.001	
1	3.84	5.02	6.63	7.88	10.83		
2	5.99	7.38	9.21	10.60	13.81		
3	7.81	9.35	11.34	12.84	16.27		
4	9.49	11.14	13.28	14.86	18.47		
5	11.07	12.83	15.09	16.75	20.52		
6	12.59	14.45	16.81	18.55	22.46		
7	14.07	16.01	18.48	20.28	24.32		
8	15.51	17.53	20.09	21.96	26.13		
9	16.92	19.02	21.67	23.59	27.88		
10	18.31	20.48	23.21	25.19	29.59		
11	19.68	21.92	24.73	26.76	31.26		
12	21.03	23.34	26.22	28.30	32.91		
13	22.36	24.74	27.69	29.82	34.53		
14	23.68	26.12	29.14	31.32	36.12		
15	25.00	27.49	30.58	32.80	37.70		
16	26.30	28.85	32.00	34.27	39.25		
17	27.59	30.19	33.41	35.72	40.79		
18	28.87	31.53	34.81	37.16	42.31		
19	30.14	32.85	36.19	38.58	43.82		
20	31.41	34.17	37.57	40.00	45.32		
21	32.67	35.48	38.93	41.40	46.80		
22	33.92	36.78	40.29	42.80	48.27		
23	35.17	38.08	41.64	44.18	49.73		
24	36.42	39.36	42.98	45.56	51.18		
25	37.65	40.65	44.31	46.93	52.62		
26	38.89	41.92	45.64	48.29	54.05		
27	40.11	43.19	46.96	49.64	55.48		
28	41.34	44.46	48.28	50.99	56.89		
29	42.56	45.72	49.59	52.34	58.30		
30	43.77	46.98	50.89	53.67	59.70		
40	55.76	59.34	63.69	66.77	73.40		
50	67.50	71.42	76.16	79.49	86.66		
60	79.08	83.30	88.38	91.95	99.61		
70	90.53	95.02	100.43	104.22	112.32		
80	101.88	106.63	112.33	116.32	124.84		
90	113.15	118.14	124.12	128.30	137.21		
100	124.34	129.56	135.81	140.17	149.44		

Formula Sheet

Standard deviation

$$s = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}}$$

χ^2 test

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

$v = c - 1$

t-test

$$t = \frac{|\bar{X}_A - \bar{X}_B|}{\sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}}}$$

$v = n_A + n_B - 2$

standard error

$$S_M = \frac{s}{\sqrt{n}}$$

- s = standard deviation
- Σ = sum of
- $\bar{X}_i$ = individual value
- $\bar{X}$ = mean of the set of values
- n = sample size
- v = degrees of freedom
- c = number of classes
- O = observed 'value'
- E = expected 'value'

Degrees of freedom (df)

Table I: Critical Values for the t-Test

Degrees of freedom (df)	Level of significance (P)						
	Two-tailed test						
	0.05	0.025	0.01	0.005	0.001	0.0005	0.001
1	6.314	12.706	31.821	63.657	318.31	636.62	31.821
2	2.920	4.303	6.965	9.925	22.327	31.598	6.965
3	2.353	3.182	4.541	5.841	10.214	12.924	4.541
4	2.132	2.776	3.747	4.604	7.173	8.610	3.747
5	2.015	2.571	3.365	4.032	5.893	6.869	3.365
6	1.943	2.447	3.143	3.707	5.208	5.959	3.143
7	1.895	2.365	2.998	3.499	4.785	5.408	2.998
8	1.860	2.306	2.869	3.355	4.501	5.041	2.869
9	1.833	2.262	2.821	3.250	4.297	4.781	2.821
10	1.812	2.228	2.764	3.169	4.144	4.587	2.764
11	1.796	2.201	2.718	3.106	4.025	4.437	2.718
12	1.782	2.179	2.681	3.055	3.930	4.318	2.681
13	1.771	2.160	2.650	3.012	3.862	4.221	2.650
14	1.761	2.145	2.624	2.977	3.787	4.140	2.624
15	1.753	2.131	2.602	2.947	3.733	4.073	2.602
16	1.746	2.120	2.583	2.921	3.686	4.015	2.583
17	1.740	2.110	2.567	2.898	3.646	3.965	2.567
18	1.734	2.101	2.552	2.878	3.610	3.922	2.552
19	1.729	2.093	2.539	2.861	3.579	3.883	2.539
20	1.725	2.086	2.528	2.845	3.552	3.850	2.528
21	1.721	2.080	2.518	2.831	3.527	3.819	2.518
22	1.717	2.074	2.508	2.819	3.505	3.792	2.508
23	1.714	2.069	2.500	2.807	3.485	3.767	2.500
24	1.711	2.064	2.492	2.797	3.467	3.745	2.492
25	1.708	2.060	2.485	2.787	3.450	3.725	2.485
26	1.706	2.056	2.479	2.779	3.435	3.707	2.479
27	1.703	2.052	2.473	2.771	3.421	3.690	2.473
28	1.701	2.048	2.467	2.763	3.408	3.674	2.467
29	1.699	2.045	2.462	2.756	3.396	3.659	2.462
30	1.697	2.042	2.457	2.750	3.385	3.646	2.457

Table I (contd): Critical Values for the t-Test

Degrees of freedom (df)	Level of significance (P)						
	Two-tailed test						
	0.05	0.025	0.01	0.005	0.001	0.0005	0.001
31	1.696	2.040	2.453	2.744	3.375	3.634	2.453
32	1.694	2.037	2.449	2.739	3.366	3.623	2.449
33	1.693	2.034	2.445	2.733	3.356	3.611	2.445
34	1.692	2.032	2.441	2.728	3.348	3.601	2.441
35	1.689	2.030	2.437	2.723	3.340	3.591	2.437
36	1.688	2.028	2.434	2.719	3.333	3.582	2.434
37	1.687	2.026	2.431	2.715	3.326	3.574	2.431
38	1.686	2.024	2.429	2.712	3.319	3.566	2.429
39	1.685	2.022	2.426	2.708	3.313	3.558	2.426
40	1.684	2.021	2.423	2.704	3.307	3.551	2.423
41	1.683	2.020	2.421	2.701	3.301	3.544	2.421
42	1.682	2.018	2.418	2.698	3.296	3.538	2.418
43	1.681	2.017	2.416	2.695	3.291	3.532	2.416
44	1.680	2.016	2.414	2.692	3.286	3.526	2.414
45	1.680	2.014	2.412	2.690	3.281	3.520	2.412
46	1.679	2.013	2.410	2.687	3.277	3.515	2.410
47	1.678	2.012	2.408	2.685	3.273	3.510	2.408
48	1.677	2.011	2.406	2.682	3.269	3.505	2.406
49	1.677	2.010	2.405	2.680	3.265	3.500	2.405
50	1.676	2.009	2.403	2.678	3.261	3.496	2.403
51	1.675	2.008	2.401	2.676	3.258	3.492	2.401
52	1.675	2.007	2.400	2.674	3.254	3.488	2.400
53	1.674	2.006	2.399	2.672	3.251	3.484	2.399
54	1.674	2.005	2.397	2.670	3.248	3.480	2.397
55	1.673	2.004	2.396	2.668	3.245	3.476	2.396
56	1.673	2.003	2.395	2.666	3.242	3.473	2.395
57	1.672	2.002	2.393	2.665	3.240	3.469	2.393
58	1.672	2.002	2.392	2.663	3.237	3.466	2.392
59	1.671	2.001	2.391	2.662	3.234	3.463	2.391
60	1.671	2.000	2.390	2.660	3.232	3.460	2.390